

Design principles to consider

- Stop guessing your capacity needs
- Test systems at production scale
- Automate to make architectural experimentation easier
- Allow for evolutionary architectures
- Drive architectures using data

also support different open source and proprietary database engines while drastically reducing operational, maintenance, scalability, and security efforts. Working on a database has been simplified through serverless computing. One good serverless database is AWS Aurora Serverless. It is cost-effective and supports both open source and proprietary software. AWS DynamoDB, Azure Cosmos DB, MongoDB and Google Cloud Big Table are also supported by serverless computing.

Block storage: Along with databases, block storages are also essential as they store media, and log files at scale. Large volumes of data, based on the type of application, can be stored by enterprises as well. With the need for storage space increasing, this serverless solution charges nominally as per usage, and is highly available and durable as well. AWS S3, Azure Blob Storage, Google Cloud Storage are some of the popular serverless block storages available.

Continuous integration and continuous deployment: Playing an important role in deployment of applications, a serverless CI/CD solution allows developers themselves to manage CI and CD (though automation is a one-time process). It saves time, is prone to fewer errors, and is readily available and cost-effective. The consistent server performance leads to a faster build process as businesses now expect quick releases. AWS Code, Pipeline, GitLab CI, GitHub Actions, Azure DevOps, CircleCI and Bamboo are a few serverless CI/CDs.

Serverless is a paradigm shift

“Serverless is not for all use cases, but for only a few that can be implemented. It can be only be a part of a solution, not the entire solution.” This is a myth and not a fact.

Serverless computing can be implemented in mobile applications by placing a simple API gateway. Real-time analytics can also be done by utilising a combination of different services.

Serverless is popular and powerful when it comes to usage. Triggering event driven platforms like SNS, S3, DynamoDB, and Lambda can result in customised performance and execution. The Lambda layer is very powerful for implementing some applications.

Due to its simplicity, serverless computing needs a few design principles (see box). It must be designed smartly by architect developers since even though each service is different, assembling all of them in the right fashion is what matters. Earlier, people would guess a capacity and design the service accordingly. However, if something did change from the capacity perspective, a huge rework had to be done in setting up the infrastructure. Instead of being given an architecture system and expecting it to survive for the next

five years, evolution of services makes experimentation easy in serverless computing. So it has given engineers and developers new parameters to think and plan accordingly. Hence, serverless computing is a paradigm shift and not just a technology.

Myths about serverless computing

1. *Serverless comes with latency:* Giving cost-effective solutions along with a provision to implement these services doesn't mean there is always a latency. There are ways to fix this, which unfortunately many are not aware of.
2. *Only compute is serverless:* Based on the landscape and catalogue of serverless, it is quite obvious that this is not the case. Many of us use serverless in various aspects but are not aware of it.
3. *All serverless APIs are microservices:* All microservices are not serverless though it can be used for the same.
4. *Serverless is only for limited use cases:* As per the different architectural flavours seen, we can conclude that this is not true.

Amidst all that has been spoken about here, the privacy of data being processed is not compromised in serverless, as all cloud providers give a provision to implement a certain way of keeping data private. AWS Cognito is both popular and powerful for data privacy. Similarly, for databases and transition of data, there are provisions to keep the written data in an encrypted way. In conclusion, serverless is a shift in mindset and not just a technology that developers and architects need to understand. **END** 🐧

By: Raghvendra Singh Dikhit

The author is senior solutions architect at Impetus Technologies. He is a technical thought leader, who is passionate about technology and technical solutions. He has experience in the IT field, mainly in the design and development of enterprise solutions. This article is based on a talk he gave at OSI 2021, a virtual event.

Transcribed and curated by Sharon Abhignya Katta